Yierpani Yimamu

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Research interests: Supply chain AI and machine learning; deep reinforcement learning for inventory and logistics; data-driven optimization and supply-chain decision making.

EDUCATION

Hefei University of Technology | Sep 2024 – Jun 2027 (expected)
Master's Candidate in Logistics Engineering and Management
Mean score: 82.60/100 | Advisor: Prof. Jingxian Chen
Management Science and Engineering: **ranked 4th in China** (ShanghaiRanking, 2025).

Selected graduate coursework: Optimization Methods (87/100); Management Research Methods (86/100); Numerical Analysis; Systems Engineering; Inventory Management; Logistics System Modeling and Simulation; Logistics System Planning and Design.

Shanghai Ocean University | Sep 2019 – Jun 2023
Bachelor of Management in Logistics Management | Mean score: 78.36/100

Selected undergraduate coursework: SAP Enterprise Resource Planning Practice (93/100); Transportation (91/100); Modern Logistics Information Management (89/100); Linear Algebra (82/100); Logistics System Design and Optimization (82/100); Supply Chain Management (81/100); International Logistics (80/100).

RESEARCH EXPERIENCE

Age-Stratified Perishable Inventory Optimization | Dec 2025 – Present
Graduate Researcher | NSFC Special Project No. 72542001

- **Develop** an age-stratified inventory MDP and digital-twin simulator in Python/PyTorch for joint replenishment and markdown decisions, incorporating multinomial-logit consumer choice and Gamma-Poisson demand.
- **Independently design** DQN/PPO states, actions, rewards and experiments; implement preference-conditioned PPO with Chebyshev scalarization and age-monotone markdown constraints.
- **Evaluated** 11 profit-waste preferences over 30 independent 200-period simulation episodes; the balanced policy reduced expired inventory by 82.01% with a 6.76% profit trade-off relative to the profit-first policy.
- **Synthesized** literature reviews, model refinements, sensitivity analyses and simulation datasets into patent documentation.

Algorithmic Collusion in Supply Chains | 2026 – Present
Graduate Researcher | NSFC General Program No. 72571084

- **Conduct** DQN/PPO numerical experiments on joint dynamic pricing and inventory under algorithmic collusion, analyzing pricing and inventory strategies through game-theoretic simulations.
- **Produce** simulation datasets, experimental figures, comparative analyses and research reports for a working paper in preparation.

Tobacco Logistics Routing and Packing Optimization | Oct 2026 – Present
Graduate Researcher | Chinese tobacco sector project

- **Analyze** approximately 100,000 sales records spanning two years and 40,000 transport records supplied by a Chinese tobacco bureau to support logistics planning.
- **Develop** vehicle routing problem (VRP) and bin-packing models to optimize delivery routes and loading plans, targeting lower distribution costs.

Chery Group Milk-Run Optimization | Summer 2025
Industry-university research project

- **Reproduced** Recursive Partitioning, Five-block and L-Algorithm components of a C++ method for packing identical rectangles; checked solutions on demonstration instances.
- **Independently developed** a Python milk-run application with vehicle/supplier inputs, time-window constraints, optimized-route visualization, cost breakdown and algorithm-iteration tracking.

PATENT APPLICATION

Co-inventor, "A Data-Driven Age-Stratified Inventory Optimization Method and System for Perishable Products."
Chinese Invention Patent Application No. 202611386809.1 | Application accepted; not granted

TECHNICAL AND QUANTITATIVE SKILLS

Programming and data analysis: Python (NumPy, pandas, Matplotlib); C++; SQL; simulation, data processing and GUI development.

Deep learning and reinforcement learning: PyTorch; Gym/Gymnasium; DQN; PPO; Markov Decision Processes; value-based learning; policy gradients and actor-critic models; state/action/reward design; preference-conditioned control.

Optimization and quantitative methods: Gurobi; CPLEX; OR-Tools; mathematical optimization; operations research; game-theoretic modeling; stochastic simulation.

Mathematical foundations: Calculus; linear algebra; probability and statistics; numerical analysis.

Research workflow: Linux; Git/GitHub; Docker; LaTeX; experimental design; sensitivity analysis; comparative and multi-metric policy evaluation.

Business analytics and enterprise tools: Power BI; Excel; SPSS for questionnaire-data analysis; SAP/ERP; PowerPoint presentation design; Word.

HONORS AND AWARDS

- **Awarded** the Second-Class Graduate Academic Scholarship, Hefei University of Technology (2026–2027).
- **Awarded** the Third-Class Graduate Academic Scholarship, Hefei University of Technology (2024–2025).

CERTIFICATIONS

English: College English Test Band 6 (CET-6), Dec 2025.

Computing: National Computer Rank Examination, Level 2 - Python Programming (Mar 2026).

Supply chain: CSCMP Supply Chain Management Certificate.

Business intelligence: Power BI Business Intelligence and Data Visualization Vocational Skills Certificate.

GITHUB PROJECTS

2D Bin Packing | C++

Open-source reproduction for packing identical rectangles: Recursive Partitioning, Five-block and L-Algorithm.
Code available: GitHub repository

Reference paper: "An effective recursive partitioning approach for the packing of identical rectangles in a rectangle"

PaperRenamer | Python GUI

- **Developed** PDF metadata extraction and Crossref/arXiv enrichment, BibTeX integration and configurable academic filename templates for literature management. Code available: GitHub repository

PROFESSIONAL EXPERIENCE

Export Sales Representative | Mar 2024 – Jun 2024

Shanghai Sangye Textile Co., Ltd. | *Full-time*

- **Managed** customer inquiries, export orders and cross-border delivery schedules; reconciled order and transport records, compiled shipment data and conducted customer outreach.

Operations Assistant Intern | Jan 2023 – Feb 2023

Shanghai Hufang Architectural Design Co., Ltd., Kashgar Branch

- **Compiled** operating reports and daily documentation, consolidated operational data and processed customer-service requests.

UNDERGRADUATE COURSE PROJECTS

SAP-ERP Supply-Chain Simulation: Executed procurement orders, managed simulated warehouse inventory and verified replenishment lead times in a SAP-ERP sandbox.

E-Commerce and Logistics Simulation: Operated an online specialty store, managed customer orders and simulated end-to-end logistics; practiced customs declarations, trade documents and commercial-invoice preparation and reconciliation.

ACADEMIC ENGAGEMENT AND SERVICE

Academic engagement: Attended the International New Energy and Intelligent Vehicle Conference, Hefei (Nov 2025), following expert reports on AI and industrial applications.

Campus leadership: Served as Class Sports Committee Member and Counselor's Assistant; coordinated sports, educational visits, reading-sharing sessions and welcome events; served as gala emcee and performer.

Community service: Completed 9 hours of administrative volunteer service in the School of Management student affairs office (Jun 2026); donated blood voluntarily (2019).

Additional credentials: Putonghua Proficiency Test, Level 2-A (Aug 2026); C1 driver's license.